NYC TAXI MLOPS PLATFORM — END-TO-END SYSTEM ARCHITECTURE

Production Inference, Model Governance, Dual Observability & Continuous Retraining

1. CLIENT & UI TIER

Frontend Presentation Layer

- Streamlit Web App (:8089)
- Auto-Distance Estimator
- Live Metric Dashboards
- Quick Route Presets
- Interactive Fare Calculator

HTTP /predict

2. MODEL SERVING

FastAPI High-Speed Inference

- FastAPI Container (:8085)
- Internal K8s DNS (:8000)
- DictVectorizer Pipeline
- Ridge @champion Predictor
- Sub-5ms Inference Latency

Loads Model

3. MODEL REGISTRY

MLflow & MinIO Governance

- MLflow Tracking (:8087)
- PostgreSQL Metadata DB
- MinIO S3 Store (:8088/:9000)
- Champion Model Alias
- Experiment Scoreboard

Drift Baseline

4. DUAL OBSERVABILITY

Data Drift & Log Telemetry

- Evidently AI Service (:8086)
- Baseline Reference Parquet
- Data Drift & Target Drift
- ELK Stack Log Forwarder
- Automated Retraining Alert

Drift Detected (>0.30)

5. CONTINUOUS INTEGRATION, DEPLOYMENT & RETRAINING (CI/CD/CT)

Closed-Loop Automation & Pipeline Orchestration

- GitHub Actions Workflow: Unit tests, Linting, Pytest validation suite on every PR & push to main branch
- Docker Multi-Platform Container Build: Multi-stage builds for serving & UI pushed to GitHub Container Registry (GHCR)
- Kubernetes Zero-Downtime Rollout: Rolling restart deploy/nyc-taxi-serving & deploy/nyc-taxi-ui with zero dropped requests
- Drift-Triggered Retraining Loop: Evidently drift detector flags data shifts -> triggers src/data_prep.py & src/train.py -> registers new Champion